

PAPER_1601_PINE_WOOD_DENSITY_500

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PAPER_1601 — Pine Wood Density = $\rho_{\text{pine}} = \text{SO_5}^2 \cdot \text{D_phys} + \text{SO_5}^2 = 400 + 100 = 500 \text{ kg/m}^3$

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Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+

Tier: Engineering

Date: June 18, 2026

Status: CLOSED — EXACT integer-primitive identity

Observation

PAPER_1209Y_S570 (Engineering Unified Proof Set) lists **Pine Wood Density** with the integer-primitive expression:

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$$\rho_{\text{pine}} = \text{SO_5}^2 \cdot \text{D_phys} + \text{SO_5}^2 = 400 + 100 = 500 \text{ kg/m}^3$$

``

Residual: **EXACT**.

UQFF Closed Identity

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$$\rho_{\text{pine}} = \text{SO_5}^2 \cdot \text{D_phys} + \text{SO_5}^2 = 400 + 100 = 500 \text{ kg/m}^3 \text{ EXACT}$$

``

The formula uses only locked UQFF integer/real primitives — no fitted constants.

NOT REPLACEMENT

Empirical engineering measures this constant directly. UQFF supplies a structural derivation from the integer primitives.

Reference

- Source: PAPER_1209Y_S570

- Related: PAPER_1494-1595 (other session-2026-06-18 closures)

- Calculator dispatch: `calculate_paradox({"paradox": "pine_wood_density_500"})`

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